

Genetic Report: Strength, Hypertrophy and Power

Sérgio F · personalised analysis

Date: 03-10-2026

Source: genotyping chip + genotype imputation ($DR2 \geq 0.8$) · GRCh38 assembly

73 SNPs in 10 functional categories · 61 polygenic scores · cross-checked with the current stack

Not medical advice. Any change to drugs or doses is for discussion with the treating physician.

Contents

Section	Page
1. Pre-check, method and disclaimer	3
2. Executive summary	4
3. Favourable and unfavourable	5
4. Fibre type, power and speed	6
5. Myofibrillar hypertrophy and lean mass	7
6. Sarcoplasmic hypertrophy and muscle energy	8
7. Strength and trainability	9
8. Muscle damage, recovery and inflammation	10
9. Tendons and connective tissue	11
10. Cardiovascular adaptation and oxygen	12
11. Glucose, fat and fuel	13
12. Brain, motivation and motor control	14
13. Hormones, mTOR and rapamycin	15
14. Integrated genetic risk profile	16
15. Convergence analysis: 4 bottlenecks	17
16. Training and rapamycin: timing (discuss with the physician)	18
17. Current management and genetic alignment: supplements	19
18. Current management and genetic alignment: medications and peptides	20
19. Gaps and multivitamin	21
20. Cross-references with prior reports	22
21. Link to the ageing mechanisms	23
22. Suggested monitoring panel	24
23. Bibliography	25

1. Pre-check, method and disclaimer

Before any interpretation, the genome build, strand orientation, call rate and imputation quality were checked. **The pre-check passed**; points with a note are flagged.

Check	Result	Status
Genome build	GRCh38 (1-based positions) confirmed for all 73 SNPs	PASS
Strand orientation	56 SNPs on the + strand with no ambiguity; 11 palindromic (A/T or C/G) assumed on the + strand, the chip output strand, with no allele conflict; 4 in – strand genes with the reading converted to the gene strand	PASS with note
Panel call rate	72 of 73 SNPs genotyped (98.6%)	PASS
Genotype origin	65 read directly on the chip; 7 imputed; 1 not found (KANSL1 rs80103986, grip-strength locus in the 17q21 region)	PASS with note
Imputation quality	DR2 0.95–1.00 for the 7 imputed SNPs; none with DR2 < 0.8	PASS
Call quality	No low-quality calls. MQ and DP are sequencing metrics and do not exist on a chip; quality comes from the chip call and DR2	Not applicable
X chromosome	FAM9B hemizygous (single allele), consistent with the donor sex	PASS
Not readable on a chip	Androgen-receptor CAG repeat; ACE insertion/deletion (rs4343 used as proxy); rare large-effect variants	Stated limit

How to read the tables

Colour	Meaning
Red	Homozygous (or hemizygous) for the risk allele; extreme unfavourable polygenic score
Yellow	Heterozygous, intermediate effect or mixed evidence; moderately unfavourable score
Green	No risk allele; typical score
Blue	Protective or favourable
Red with *	Low quality (DR2 < 0.8 or score coverage < 80%); not used for decisions
Grey	SNP not found in the data: treated as likely homozygous reference

- † = frequency or effect-size data weak or absent in the reference population (candidate genes studied in small samples). These SNPs do not drive recommendations.
- Polygenic-score **percentiles** are calculated against a reference population of 633 genomes. Consensus = median of the scores with ≥ 80% coverage.
- **Genotypes** on the GRCh38 + strand, with the usual variant name; "in the gene" marks the strand used in the literature.

Disclaimer

This report interprets genetic data from a genotyping chip with imputation, **not from whole-genome sequencing**. Missing SNPs are stated explicitly and treated as likely homozygous reference, with the limited confidence this implies on a chip. The effects of almost all strength and hypertrophy genes are small and the evidence for several is mixed (especially **ACTN3** and **ACE**). Polygenic scores show tendencies, not destinies. Everything about medication is **information for the conversation with the prescribing physician**; sirolimus is a narrow-therapeutic-index immunosuppressant, and any pharmacogenomic finding is informational only. Every change to the stack appears as a **suggestion (not applied)**: the real stack stays as it is.

2. Executive summary

In short: strength is normally trainable and the aerobic engine and recovery are favourable, but **baseline lean mass is genetically low, tendon is the weak link** and **spontaneous drive to train is low**. Hypertrophy will need more volume, more consistency and slower load progression for tendons than for muscle.

Primary vulnerabilities (homozygous risk and extreme scores)

- **COL5A1 rs12722 T/T** (homozygous risk): stiffer tendons and more Achilles tendinopathy. Moderate evidence.
- **Low polygenic lean mass:** appendicular **p9.5**, whole body **p28** (consensus), trunk p8–34; genetic IGF-1 **p32**.
- **Genomic body-fat percentage p92**, with risk alleles in FTO, TCF7L2 and MTNR1B.
- **Low polygenic physical activity:** measured p14, self-reported p7, gym attendance p12.6.
- **FAM9B hemizygous T** (slightly lower testosterone): common (~72% of men) with a small effect.

Secondary findings (heterozygous or moderate)

- **ACTN3 RX** and probable **ACE I/D**: intermediate, mixed evidence. **AGT M/T**, **PPARA G/C**.
- **Genetically high SHBG (consensus p83)**: three higher-SHBG alleles; less free testosterone in proportion to the total (genetic total p70).
- **Nitric oxide axis:** NOS3 Glu298Asp and –786 T/C, EDN1 Lys198Asn and AGTR1 A/C, all heterozygous.
- **LILRB5 Asp/Gly** (higher CK, statin myalgia), **PPARD A/G** (smaller metabolic training response), **MPP7** (1 Achilles-injury allele), **SOD2 Ala/Ala** (mixed).
- **HFE H63D** heterozygous with genetic ferritin p80; **BDNF Val/Met**; **CYP3A5 *1/*3** expresser (less sirolimus per mg; informational).

Genetically protected pathways

- **Mitochondrial engine:** PPARGC1A Gly/Gly, normal AMPD1, typical TFAM.
- **Recovery and inflammation:** MYLK without damage alleles, TNF –308 G/G, CCL2 A/A, genetic CRP p26.
- **Glucose:** genetic HbA1c p7.5, MC4R T/T, ADRB3 Trp/Trp.
- **Typical grip strength** (consensus p51); IL15RA A/A and TRHR with 1 lean-mass allele; typical myostatin and follistatin.
- **Drugs and longevity:** SLCO1B1 Val/Val (no genetic statin-myopathy risk), FOXO3 G/G (homozygous for longevity), HFE without C282Y, genetic 25-OH-D p85.

3. Favourable and unfavourable

The 5 most favourable findings

#	Finding	What it means	Evidence
1	PPARGC1A Gly/Gly + normal AMPD1	Good mitochondrial and capillary base, good tolerance of high volume and intact rapid ATP regeneration.	Moderate
2	Favourable recovery: TNF -308 G/G, CCL2 A/A, MYLK without damage alleles, genetic CRP p26	Less inflammation and damage after hard sessions; tolerates a higher training frequency.	Moderate to weak
3	Protected glucose: genetic HbA1c p7.5, MC4R T/T, ADRB3 Trp/Trp	A good glycaemic base for using muscle as a glucose sink.	Strong
4	Trainable strength: grip p51, ACTN3 RX (functional fast fibres)	No genetic limit on strength or power.	Moderate (ACTN3: mixed)
5	SLC01B1 Val/Val + FOXO3 G/G	No genetic statin-myopathy risk; a longevity profile on the FOXO axis.	Strong

The 5 most unfavourable findings

#	Finding	What it means	Evidence
1	COL5A1 rs12722 T/T	Stiffer tendon and more tendinopathy: tendon adapts more slowly than muscle.	Moderate
2	Low lean mass: appendicular p9.5, whole body p28; IGF-1 p32	Gaining mass needs more volume, more protein and more consistency than average.	Strong (GWAS/PGS)
3	Genomic body-fat percentage p92 (+ FTO, TCF7L2, MTNR1B)	A tendency to store fat; muscle is the main tool to counter it.	Strong
4	Low drive to train: measured activity p14, self-reported p7, gym p12.6; BDNF Val/Met	Consistency depends on structure (fixed times, goals, company), not willpower.	Moderate
5	High SHBG (p83) + FAM9B T	Less free testosterone in proportion to the total; a slightly smaller baseline anabolic signal.	Strong (small effect)

4. Fibre type, power and speed

Power depends on fast (type II) fibres, which contract quickly and generate high force in little time. These genes affect the structure of those fibres, rapid ATP regeneration and the balance between fast and slow fibres.

Gene	SNP · variant	Genotype	Origin	Effect allele	Reading	Evidence
ACTN3	rs1815739 · R577X (c.1729C>T)	TC · RX (one functional copy)	chip	T (X, stop codon)	One functional α -actinin-3 copy: fast fibres preserved	Mixed †
AMPD1	rs17602729 · C34T (Q12X)	GG · normal (no Q12X)	chip	A (34T, stop codon)	Normal AMPD1: rapid ATP regeneration intact	Moderate
GALNTL6	rs558129 · intronic	GA · C/T in the gene (1 T)	chip	T (gene strand)	1 T allele (power; small study)	Weak †
ACE	rs4343 · A2350G (I/D proxy; G tags D)	GA · G/A (probable I/D)	chip	G (tags the D allele)	Probable I/D: intermediate ACE activity	Mixed †
AGT	rs699 · M235T (c.704T>C)	AG · M/T at codon 235	chip	G (235T)	M/T: intermediate	Weak †
HIF1A	rs11549465 · P582S (C1772T)	CC · Pro/Pro	chip	T (Ser582, advantage)	Pro/Pro: no advantage allele; typical	Weak †
PPARA	rs4253778 · intron 7 G/C	GC · G/C	chip	C (power)	G/C: 1 power-associated allele	Mixed †
IL6	rs1800795 · -174 G>C	GG · -174 G/G	chip	G (-174G)	G/G: no demonstrated training effect	Mixed †

ACE: the Alu insertion/deletion cannot be read on a chip; rs4343 is a strongly linked proxy marker. The evidence for ACTN3 and ACE in strength and power is **mixed**: replicated in elite athletes, inconsistent in non-athletes (Ma 2013; Puthuchear 2011; Clarkson 2005).

*A **mixed** fibre profile. **ACTN3 RX** has one functional copy of α -actinin-3, enough for fast fibres (RX and RR behave similarly in most studies), and lacks the XX genotype linked to more muscle damage. Probable **ACE I/D** and **AGT M/T** are intermediate. **AMPD1** is normal. **HIF1A Pro/Pro**, **PPARA G/C** and **IL6 -174 G/G** give no demonstrated advantage or disadvantage. **Reading**: trainable power with no marked genetic leaning; the evidence for these genes is mixed.*

5. Myofibrillar hypertrophy and lean mass

Myofibrillar hypertrophy is the growth of contractile proteins (actin, myosin). The engine is mTORC1, activated by mechanical load, IGF-1/insulin and amino acids; myostatin is the brake. Lean mass is highly polygenic: polygenic scores say more than any single SNP.

Gene	SNP · variant	Genotype	Origin	Effect allele	Reading	Evidence
IGF1	rs35767 · promoter -383 C>T	GG · C/C in the gene (no T)	chip	T in the gene (A on + strand)	No higher-IGF-1 T allele	Moderate
IGF1R	rs2229765 · E1043E (G3174A)	GA · G/A	chip	A	G/A: uncertain effect	Weak ↑
TRHR	rs7832552 · intronic	CT · C/T (1 more-lean-mass allele)	chip	T (more lean mass)	1 more-lean-mass allele	Moderate
MSTN	rs1805086 · K153R (c.458A>G)	TT · K/K (the common type)	chip	C (153R)	K/K: typical myostatin	Weak ↑
FST	rs3797297 · intronic	GG · G/G (no T)	imputed (DR2 0.95)	T	No follistatin variant	Weak ↑

Polygenic score	ID	Percentile	Coverage	Direction
Whole-body fat-free mass	PGS003531	p29.5	99.0%	higher = better
Whole-body fat-free mass	PGS002168	p27.0	97.1%	higher = better
Whole-body fat-free mass	PGS004376	p32.4	97.0%	higher = better
Whole-body fat-free mass	PGS003901	p3.5	93.0%	higher = better
Appendicular lean mass	PGS005315	p9.5	89.6%	higher = better
Trunk fat-free mass	PGS004403	p34.0	97.0%	higher = better
Trunk fat-free mass	PGS003923	p8.2	93.0%	higher = better
Arm fat-free mass	PGS004395	p25.3	97.0%	higher = better
Leg fat-free mass	PGS004387	p35.7	97.0%	higher = better
IGF-1	PGS002178	p28.8	97.5%	higher = better
IGF-1	PGS003539	p26.4	99.1%	higher = better
IGF-1	PGS001960	p32.1	96.3%	higher = better
IGF-1	PGS004338	p75.7	95.9%	higher = better
IGF-1	PGS000687	p34.1	90.0%	higher = better

This is the **weakest point** of the profile. Lean-mass polygenic scores are below average (whole body **p28**; appendicular **p9.5**; trunk **p8–34**; arms **p25**; legs **p36**) and genetic IGF-1 is low-typical (**p32**), consistent with IGF1 rs35767 without the T allele. TRHR carries 1 more-lean-mass allele; myostatin is K/K (the common type) and follistatin has no variant. **Reading:** gaining mass will take more volume, more protein and more consistency than average. This does not prevent gains; it changes the training dose needed.

6. Sarcoplasmic hypertrophy and muscle energy

Sarcoplasmic hypertrophy is expansion of the cytoplasm (glycogen, enzymes, mitochondria, water) without a proportional rise in myofibrils. It is real but hard to measure, and **no validated SNP specific to it exists**; genes of muscle energy metabolism are used as a proxy.

Gene	SNP · variant	Genotype	Origin	Effect allele	Reading	Evidence
PPARGC1A	rs8192678 · G482S (c.1444G>A)	CC · Gly/Gly	chip	T (Ser482)	Gly/Gly: fully active PGC-1α (favourable)	Moderate
TFAM	rs1937 · S12T	GG · Ser/Ser	chip	C (Thr12)	Ser/Ser: typical	Weak †
SLC16A1	rs1049434 · D490E (T1470A)	AT · heterozygous	chip	T (gene strand, 1470T)	Heterozygous: intermediate lactate transport	Weak †
CKM	rs8111989 · 3' UTR NcoI A/G	TT · A/A in the gene	chip	A/G (NcoI; no consensus risk allele)	A/A in the gene: no clear meaning	Mixed †

PPARGC1A Gly/Gly is favourable: fully active PGC-1α, a good mitochondrial and capillary base, and good tolerance of volume. **TFAM Ser/Ser** is typical. The **MCT1** lactate transporter is heterozygous (intermediate transport). **CK-MM** is A/A in the gene, with no clear meaning. There are no validated SNPs specific to sarcoplasmic hypertrophy; the good mitochondrial engine suggests good tolerance of longer sets and high volume.

7. Strength and trainability

Trainability (how much is gained from the same training) varies widely: with strength training, size gains range from almost zero to over 50%. Part of this variation runs in families, but almost no single gene predicts it well; strength GWAS loci have small effects that add up.

Gene	SNP · variant	Genotype	Origin	Effect allele	Reading	Evidence
CNTF	rs1800169 · G-A (null allele, splicing)	GG · G/G (normal CNTF)	chip	A (null allele)	Normal CNTF	Weak ↑
IL15RA	rs2296135 · 3' UTR A/C	AA · A/A	imputed (DR2 0.96)	A (more hypertrophy)	A/A: higher-hypertrophy genotype (small study; not used for decisions)	Weak ↑
PPARD	rs2267668 · intronic A>G	GA · A/G (1 G)	chip	G (lower response)	1 allele of a smaller metabolic training response	Weak ↑
ACSL1	rs6552828 · intronic	GG · G/G (no A)	chip	A (+ strand)	No A allele	Weak ↑
TGFA	rs958685 · grip GWAS locus	AC · 1 increasing allele	imputed (DR2 0.99)	A (more strength)	1 increasing allele (of 2)	Moderate
POLD3	rs72979233 · grip GWAS locus	AG · 1 increasing allele	chip	A (more strength)	1 increasing allele (of 2)	Moderate
HOXB3	rs2288278 · grip GWAS locus	GG · 0 increasing alleles	chip	A (more strength)	0 increasing alleles (of 2)	Moderate
GLIS1	rs4926611 · grip GWAS locus	CC · 2 increasing alleles	imputed (DR2 0.98)	C (more strength)	2 increasing alleles (of 2)	Moderate
PEX14	rs6687430 · grip GWAS locus	GA · 1 increasing allele	chip	G (more strength)	1 increasing allele (of 2)	Moderate
KANSL1	rs80103986 · grip GWAS locus	no data	not found	A (more strength)	Not found in the data (17q21 region, hard to impute): treated as likely homozygous reference	Moderate
LRPPRC	rs10186876 · grip GWAS locus	GG · 0 increasing alleles	chip	A (more strength)	0 increasing alleles (of 2)	Moderate

Grip loci: blue = 2 increasing alleles, green = 1, yellow = 0, grey = not found. Per-allele effect $\approx 0.14\text{--}0.18$ strength units; they only matter when summed (Willems 2017).

Polygenic score	ID	Percentile	Coverage	Direction
Grip strength	PGS003513	p51.3	98.8%	higher = better
Grip strength	PGS002143	p52.4	96.8%	higher = better
Grip strength	PGS001927	p29.5	96.3%	higher = better
Usual walking pace	PGS001075	p57.5	90.2%	higher = better
General frailty factor	PGS005221	p74.6	89.9%	higher = worse

Polygenic grip strength is **typical** (consensus **p51** of three scores: 51.3, 52.4 and 29.5). At the genotyped grip GWAS loci there are 5 increasing alleles out of 12 possible (≈ 6.8 expected in the reference population), slightly below; KANSL1 rs80103986 **was not found in the data** (17q21 inversion region, hard to impute) and is treated as likely homozygous reference. CNTF is normal. IL15RA A/A is the genotype linked to more hypertrophy in one small study (weak evidence, not used for decisions). PPARD A/G carries one allele of the smaller metabolic training response. **Reading:** strength is normally trainable; the metabolic response to aerobic training may be slightly smaller.

8. Muscle damage, recovery and inflammation

Eccentric exercise causes micro-injury; recovery involves controlled inflammation (macrophages, cytokines) and satellite cells. These genes affect damage, markers (CK) and the inflammatory response, and include the pharmacogenomics of statin myopathy.

Gene	SNP · variant	Genotype	Origin	Effect allele	Reading	Evidence
MYLK	rs28497577 · C37885A (P21H)	GG · C/C in the gene (no 37885A)	chip	A in the gene (37885A)	No 37885A damage allele	Weak †
MYLK	rs2700352 · C49T (P147S)	GG · C/C in the gene (no 49T)	imputed (DR2 0.99)	T in the gene (49T)	No 49T damage allele	Weak †
LILRB5	rs12975366 · D247G (T = Asp247)	TC · Asp/Gly	chip	T (Asp247)	1 Asp247 allele: higher resting CK and statin myalgia	Moderate
TNF	rs1800629 · -308 G>A	GG · -308 G/G	chip	A (-308A)	-308 G/G: no higher-TNF allele	Mixed †
CCL2	rs1024611 · -2518 A>G	AA · -2518 A/A	chip	G (-2518G)	A/A: no higher-CCL2 allele	Weak †
SOD2	rs4880 · V16A	GG · Ala/Ala	chip	G (Ala16)	Ala/Ala: more H ₂ O ₂ (adaptive signal and oxidative stress)	Mixed †
SLCO1B1	rs4149056 · V174A (c.521T>C)	TT · Val/Val (no *5/*15)	chip	C (Ala174)	Val/Val: no genetic statin-myopathy risk	Strong

Polygenic score	ID	Percentile	Coverage	Direction
CRP	PGS002873	p25.8	96.5%	higher = worse
CRP	PGS001946	p12.0	96.4%	higher = worse
CRP	PGS000314	p25.9	99.4%	higher = worse

Recovery is **genetically favourable**: MYLK without the damage alleles, TNF -308 G/G, CCL2 -2518 A/A and genetic CRP at p26. SOD2 Ala/Ala produces more H₂O₂ (an adaptive signal, but also more oxidative stress; mixed evidence). Two points of attention: **LILRB5 Asp/Gly** (one allele of higher resting CK and of statin myalgia) and ACTN3 RX (see category 1). **SLCO1B1 Val/Val** carries none of the myopathy risk alleles: with pitavastatin there is no increased genetic risk.

9. Tendons and connective tissue

Tendon adapts to load more slowly than muscle (months rather than weeks). Collagen synthesis needs vitamin C, iron and 2-oxoglutarate (hydroxylases) and copper (lysyl oxidase), and responds to mTORC1.

Gene	SNP · variant	Genotype	Origin	Effect allele	Reading	Evidence
COL5A1	rs12722 · BstUI 3' UTR C>T	TT · T/T (homozygous risk)	chip	T	T/T homozygous risk: stiffer tendon, more tendinopathy	Moderate
COL1A1	rs1800012 · Sp1 G>T (T allele = "s")	CC · G/G in the gene (SS, no s allele)	chip	T in the gene ("s" allele, A on + strand)	No s allele: no rupture protection; favourable for bone	Mixed †
COL3A1	rs1800255 · A698T	GG · G/G (no A)	chip	A (698T)	No risk allele	Weak †
GDF5	rs143383 · +104T/C 5' UTR	GG · C/C in the gene (no T)	chip	T in the gene (A on + strand)	No risk allele	Moderate
MMP3	rs679620 · E45K	CC · C/C	chip	E45K (no consensus)	C/C: no consensus on the effect	Weak †
MPP7	rs1249269 · Achilles injury GWAS locus	TC · C/T (1 T)	chip	T	1 Achilles-injury risk allele (GWAS)	Moderate

Polygenic score	ID	Percentile	Coverage	Direction
Muscle, ligament and fascia disorders	PGS019710	p34.1	94.3%	higher = worse
Hip osteoarthritis	PGS002763	p38.9	97.0%	higher = worse
Knee osteoarthritis	PGS002767	p56.1	97.0%	higher = worse
Hip osteoarthritis	PGS018417	p11.4 *	34.0%	higher = worse
Knee osteoarthritis	PGS018416	p6.0 *	43.7%	higher = worse

* Coverage < 80%: low quality, not used for decisions.

The **main unfavourable finding** of the report: **COL5A1 rs12722 T/T** (homozygous risk), linked to stiffer tendons, reduced range of motion and more Achilles tendinopathy. **COL1A1** lacks the protective s allele (G/G in the gene): no protection against ruptures, but favourable for bone. **MPP7** carries one risk allele (GWAS). **GDF5** and **COL3A1** carry no risk alleles and **MMP3** is C/C. **Reading:** tendon is the weak link in the strength chain; load should progress more slowly for tendons than for muscle.

10. Cardiovascular adaptation and oxygen

Strength and power training also depend on blood flow (exercise hyperaemia), capillarisation and oxygen transport. Endothelial nitric oxide (eNOS) is the central vasodilator.

Gene	SNP · variant	Genotype	Origin	Effect allele	Reading	Evidence
NOS3	rs1799983 · Glu298Asp (G894T)	TG · Glu/Asp	imputed (DR2 1.00)	T (Asp298)	Glu/Asp: slightly less stable eNOS	Moderate
NOS3	rs2070744 · -786 T>C	CT · -786 T/C	chip	C (-786C)	-786 T/C: lower eNOS transcription	Mixed †
EDN1	rs5370 · Lys198Asn	GT · Lys/Asn	chip	T (Asn198)	Lys/Asn: 1 Asn198 allele	Weak †
VEGFA	rs2010963 · -634 G>C	GG · -634 G/G	chip	C (-634C)	G/G: typical	Weak †
ADRB1	rs1801253 · Arg389Gly	CC · Arg/Arg	chip	G (Gly389)	Arg/Arg: full cardiac catecholamine response	Weak †
AGTR1	rs5186 · A1166C	AC · A/C	chip	C (1166C)	A/C: 1 1166C allele	Mixed †
HFE	rs1799945 · H63D	CG · H63D heterozygous	chip	G (63D)	H63D heterozygous: slightly higher iron absorption	Moderate
HFE	rs1800562 · C282Y	GG · C282Y non-carrier	chip	A (282Y)	No C282Y	Strong

Polygenic score	ID	Percentile	Coverage	Direction
Haemoglobin	PGS000093	p28.0	91.5%	no judgement
Haemoglobin	PGS000168	p11.7	83.1%	no judgement
Haematocrit	PGS000167	p13.3	84.3%	no judgement
Ferritin	PGS003426	p79.8	96.4%	higher = worse
Iron metabolism disorders	PGS001823	p76.0	98.2%	higher = worse
Average heart rate	PGS001375	p32.4	88.7%	higher = worse
Heart rate	PGS000300	p4.4	94.8%	higher = worse
Heart rate	PGS003472	p58.1	97.2%	higher = worse

The nitric oxide axis has **three heterozygous variants**: NOS3 Glu298Asp, NOS3 -786 T/C and EDN1 Lys198Asn, plus AGTR1 1166 A/C. This suggests slightly lower endothelial vasodilation during exercise, which aerobic training compensates. VEGFA -634 G/G is typical and ADRB1 Arg/Arg gives a full cardiac catecholamine response. **HFE H63D** heterozygous, without C282Y, absorbs slightly more iron; genetic ferritin is at p80, so keep **ferritin <100**. Genetic haemoglobin and haematocrit are low-typical (p20 and p13).

11. Glucose, fat and fuel

Muscle is the largest consumer of glucose after a meal; more muscle and more training improve insulin sensitivity. These genes affect insulin secretion, adiposity and fat mobilisation.

Gene	SNP · variant	Genotype	Origin	Effect allele	Reading	Evidence
TCF7L2	rs7903146 · intronic C>T	CT · C/T	chip	T	1 T allele: lower insulin secretion	Strong
MTNR1B	rs10830963 · intronic C>G	CG · C/G	chip	G	1 G allele: melatonin weighs more on glucose control	Strong
PPARG	rs1801282 · Pro12Ala	CC · Pro/Pro (no Ala)	chip	G (Ala12, protective)	Pro/Pro: no protective Ala	Strong
FTO	rs9939609 · intronic T>A	TA · T/A	chip	A	1 A allele: adiposity, attenuated by physical activity	Strong
MC4R	rs17782313 · intergenic T>C	TT · T/T	chip	C	T/T: no risk allele	Strong
ADRB3	rs4994 · Trp64Arg	AA · Trp/Trp	chip	G (Arg64)	Trp/Trp: no risk allele	Mixed †
UCP2	rs660339 · A55V	GA · Ala/Val	chip	A (Val55, + strand)	Ala/Val: uncertain effect	Weak †

Polygenic score	ID	Percentile	Coverage	Direction
Body-fat percentage	PGS004374	p92.3	97.0%	higher = worse
HbA1c	PGS001953	p6.6	96.2%	higher = worse
HbA1c	PGS002697	p8.4	97.2%	higher = worse
Fasting insulin	PGS000308	p4.9	100.0%	higher = worse
Fasting insulin	PGS005276	p74.4	95.4%	higher = worse
Type 2 diabetes	PGS012547	p57.8	98.1%	higher = worse
Type 2 diabetes	PGS003443	p68.9	96.6%	higher = worse
Type 2 diabetes	PGS005335	p49.3	93.1%	higher = worse
Triglycerides	PGS002719	p68.2	97.2%	higher = worse
ApoB	PGS002102	p46.1	97.6%	higher = worse

There is one risk allele in **TCF7L2** (lower insulin secretion), **FTO** (adiposity, attenuated by physical activity) and **MTNR1B**. **PPARG** is Pro/Pro (no protective Ala). Favourable: **MC4R T/T**, **ADRB3 Trp/Trp** and **low genetic HbA1c (p7.5)**. Unfavourable: **genomic body-fat percentage at p92**. **Reading:** blood glucose is genetically protected, but there is a tendency to store fat; muscle is the main tool to counter it.

12. Brain, motivation and motor control

Early strength gains come mostly from the nervous system (recruitment and firing rate). The drive to exercise is also heritable: physical-activity GWAS find loci linked to dopamine and personality.

Gene	SNP · variant	Genotype	Origin	Effect allele	Reading	Evidence
BDNF	rs6265 · Val66Met	CT · Val/Met	chip	T (Met66)	Val/Met: less activity-dependent BDNF release	Moderate
COMT	rs4680 · Val158Met	GG · Val/Val	chip	A (Met158)	Val/Val: fast dopamine clearance; stress tolerance	Moderate
ANKK1	rs1800497 · Taq1A (Glu713Lys)	GG · A2/A2	chip	A (A1)	A2/A2: normal D2 receptors	Moderate
CADM2	rs62253088 · strenuous-sport GWAS locus	CC · C/C (no T)	imputed (DR2 0.99)	T (more strenuous sport)	No more-strenuous-sport allele	Moderate

Polygenic score	ID	Percentile	Coverage	Direction
Measured physical activity	PGS002255	p14.2	96.1%	higher = better
Self-reported physical activity	PGS002254	p7.4	96.2%	higher = better
Regular gym attendance	PGS001019	p12.6	88.9%	higher = better
Major depression	PGS004591	p85.2	100.0%	higher = worse
Major depression	PGS003576	p66.0	96.0%	higher = worse

BDNF Val/Met releases less BDNF with activity (slightly slower motor learning, which regular exercise compensates). **COMT Val/Val** clears dopamine quickly (a stress-tolerance profile). **ANKK1** is A2/A2 (normal D2 receptors) and **CADM2 C/C** lacks the more-strenuous-sport allele. Scores: **measured physical activity p14, self-reported p7, gym attendance p12.6.**

Reading: intrinsic drive to train is genetically low; fixed schedules, measurable goals and company matter more than willpower.

13. Hormones, mTOR and rapamycin

Free testosterone (not bound to SHBG) acts on the androgen receptor and raises protein synthesis and satellite cells. The receptor CAG repeat length (fewer repeats = more sensitive receptor) **cannot be read on a SNP chip**. Rapamycin inhibits mTORC1; its metabolism depends on CYP3A5/CYP3A4 and P-glycoprotein.

Gene	SNP · variant	Genotype	Origin	Effect allele	Reading	Evidence
SHBG	rs12150660 · intronic (higher SHBG)	GT · G/T (1 T)	chip	T (higher SHBG)	1 higher-SHBG allele	Strong
SHBG	rs6258 · P156L	CC · C/C	chip	T (156L, low SHBG)	C/C: typical	Strong
SHBG	rs727428 · 3' (T = more bioavailable testosterone)	TC · T/C (1 C)	chip	T (more bioavailable testosterone)	1 C allele: higher SHBG	Strong
SHBG	rs1799941 · -68 G>A (5' UTR)	GA · G/A (1 A)	chip	A (higher SHBG)	1 A allele: higher SHBG	Strong
FAM9B	rs5934505 · X locus (low testosterone)	T · T (hemizygous, X chromosome)	chip	T (low testosterone)	Hemizygous T: slightly lower testosterone; common (~72% of men)	Strong
SRD5A2	rs523349 · V89L	CC · Val/Val	chip	Leu89 (G on + strand)	Val/Val: typical	Mixed †
VDR	rs2228570 · FokI (M1T)	AG · Ff	chip	A (f allele, + strand)	Ff: intermediate	Mixed †
GC	rs2282679 · intronic (DBP)	TT · T/T (no G)	chip	G (lower 25-OH-D)	T/T: no lower-25-OH-D allele	Strong
MTOR	rs2295080 · promoter G>T	GT · G/T	chip	T/G (no consensus)	G/T: no clear meaning	Weak †
FOXO3	rs2802292 · intronic T>G (G = longevity allele)	GG · G/G (homozygous longevity)	chip	G (longevity, protective)	G/G: homozygous for longevity	Strong
CYP3A5	rs776746 · *3 (6986A>G)	CT · *1/*3 (expresser)	chip	C (*3, non-expresser)	*1/*3 expresser: less sirolimus per mg (informational)	Strong
CYP3A4	rs35599367 · *22 (intron 6 C>T)	GG · *1/*1 (normal)	chip	A (*22)	*1/*1: normal CYP3A4	Moderate
ABCB1	rs1045642 · C3435T	AA · 3435 T/T	chip	A (3435T)	3435 T/T: informational (P-gp)	Mixed †

The androgen-receptor CAG repeat (testosterone sensitivity) cannot be read on a chip: it needs long-read sequencing or fragment analysis.

Polygenic score	ID	Percentile	Coverage	Direction
SHBG	PGS003552	p87.2	99.3%	higher = worse
SHBG	PGS001977	p82.6	97.1%	higher = worse
SHBG	PGS000694	p65.7	88.4%	higher = worse
Testosterone (men)	PGS003559	p89.3	99.0%	higher = better
Testosterone (men)	PGS002205	p49.0	97.3%	higher = better
Testosterone (men)	PGS001988	p58.0	96.8%	higher = better
Testosterone	PGS000696	p69.8	89.4%	higher = better
Testosterone (men)	PGS000323	p82.9	85.8%	higher = better
25-OH vitamin D	PGS000882	p84.7	96.9%	higher = better
25-OH vitamin D	PGS019960	p91.2	87.9%	higher = better
Vitamin D	PGS002721	p84.0	97.2%	higher = better

SHBG is genetically high (consensus p83): rs12150660, rs1799941 and rs727428 each carry one higher-SHBG allele; rs6258 is C/C. Genetic total testosterone is at p70 (high SHBG raises the total but lowers the free fraction). **FAM9B** carries the low-testosterone T allele (hemizygous; present in ~72% of men). **SRD5A2** is Val/Val. **The androgen-receptor CAG repeat cannot be read on the chip.** Vitamin D: VDR Ff, GC T/T and genetic 25-OH-D at p85. MTOR G/T has no clear meaning; **FOXO3 G/G** is homozygous for longevity (protective). Pharmacogenomics: **CYP3A5 *1/*3 = expresser**, clearing more sirolimus per mg (lower exposure); CYP3A4 *1/*1; ABCB1 3435 T/T (informational).

14. Integrated genetic risk profile

Summary by category. The level describes the genetic effect on strength, hypertrophy and power, not a disease risk.

Category	Key findings	Level	Evidence
1. Fibre type, power and speed	ACTN3 RX, probable ACE I/D, AGT M/T, normal AMPD1	TYPICAL	Mixed
2. Myofibrillar hypertrophy and lean mass	Lean mass p9.5–28; IGF-1 p32; IGF1 without the T allele; TRHR 1 favourable allele	HIGH	Strong (PGS)
3. Sarcoplasmic hypertrophy and muscle energy	PPARGC1A Gly/Gly; typical TFAM; heterozygous MCT1	FAVOURABLE	Moderate
4. Strength and trainability	Grip p51; 5 of 12 increasing alleles; PPARD A/G; IL15RA A/A; KANSL1 missing	TYPICAL	Moderate
5. Muscle damage, recovery and inflammation	Favourable TNF, CCL2 and MYLK; CRP p26; LILRB5 Asp/Gly; SOD2 Ala/Ala	LOW	Moderate to weak
6. Tendons and connective tissue	COL5A1 T/T; COL1A1 without the s allele; MPP7 1 allele; muscle/ligament disorders p34	HIGH	Moderate
7. Cardiovascular adaptation and oxygen	NOS3 ×2, EDN1 and AGTR1 heterozygous; HFE H63D; genetic ferritin p80	MODERATE	Moderate to mixed
8. Glucose, fat and fuel	Body fat p92; FTO, TCF7L2, MTNR1B heterozygous; HbA1c p7.5; favourable MC4R and ADRB3	MODERATE	Strong
9. Brain, motivation and motor control	Physical activity p7–14; gym p12.6; BDNF Val/Met; COMT Val/Val	HIGH	Moderate
10. Hormones, mTOR and rapamycin	SHBG p83; total testosterone p70; FAM9B T; FOXO3 G/G; CYP3A5 expresser	MODERATE	Strong (small effects)

15. Convergence analysis: 4 bottlenecks

Points where several variants, scores and current drugs add up in the same direction.

1. Muscle that tolerates volume, tendon that adapts slowly

Variant chain	Mechanism	Clinical implication
COL5A1 T/T → COL1A1 without the s allele → MPP7 (1 allele) + PPARGC1A Gly/Gly and favourable recovery; drugs with a tendon or CK signal (isotretinoin, bempedoic acid).	Muscle recovers quickly and tolerates high volume, but tendon collagen is stiffer and adapts more slowly. The mismatch between a growing muscle and a tendon that has not yet adapted is the classic cause of tendinopathy.	Slower load progression in exercises that load the Achilles and patellar tendons; collagen with vitamin C before loading; new tendon pain is a reason to see the physician, especially on isotretinoin or bempedoic acid. Avoid fluoroquinolones when an alternative exists (discuss with the physician).

2. Low baseline anabolic signal, with two pharmacological brakes

Variant chain	Mechanism	Clinical implication
Lean mass p9.5–28 → IGF-1 p32 (IGF1 without the T allele) → SHBG p83 + FAM9B T (less free testosterone) + rapamycin (inhibits mTORC1) + retatrutide (energy deficit).	mTORC1 receives less signal from IGF-1 and free testosterone. On top of that, rapamycin cuts contraction-induced protein synthesis in the days after a dose (Drummond 2009) and an energy deficit reduces lean-mass gains (Murphy 2022).	This is the most important bottleneck for hypertrophy. Distributed protein, creatine, enough volume (Schoenfeld 2017), training timed around rapamycin and DEXA to track lean mass on retatrutide; the pace of the deficit and the rapamycin schedule are for discussion with the physician.

3. A tendency to store fat with protected glucose

Variant chain	Mechanism	Clinical implication
Body fat p92 → FTO (1 allele) → TCF7L2 (1 allele) → MTNR1B (1 allele) → PPARG without the protective Ala; offset by HbA1c p7.5, MC4R T/T and ADRB3 Trp/Trp.	The beta cell has less reserve and melatonin weighs more on insulin secretion (MTNR1B), while FTO pushes towards adiposity. Muscle is the body's largest glucose sink, and physical activity attenuates the FTO effect (Kilpeläinen 2011).	Muscle mass is a metabolic tool here, not only an aesthetic one. More fibre is always better; melatonin only 0.3–1 mg and ≥ 3 h after the last meal; larger meals close to training.

4. Little spontaneous drive and slower motor plasticity

Variant chain	Mechanism	Clinical implication
Physical activity p7–14 → gym p12.6 → CADM2 without the strenuous-sport allele → BDNF Val/Met.	The genetic drive to seek exercise is low and activity-dependent BDNF release is smaller, making motor learning slightly slower. Regular exercise raises BDNF and offsets part of the effect.	Consistency is the factor that weighs most and depends on structure: fixed times, measured goals (strength, DEXA), training with company and repeated technique before raising load. Exercise also protects mood, which matters given the depression predisposition.

16. Training and rapamycin: timing (discuss with the physician)

Rapamycin (sirolimus) inhibits mTORC1, the main switch of muscle protein synthesis. In humans, a single dose before training blocked the contraction-induced rise in protein synthesis (Drummond 2009). The half-life is ~62 h (Mahalati 2001): with 4 mg every 6 weeks, relevant exposure is concentrated in the first 2–3 days after the dose, and about 5 of the 6 weeks are free. In a one-year trial of low-dose intermittent rapamycin, lean mass was not harmed (Moel 2025).

Timing relative to the dose	Suggested training (suggestion, not applied)	Why
Before the dose	Last heavy hypertrophy session $\geq$ 24–48 h before the dose	Protein synthesis from a session lasts 24–48 h; this way it is not cut short
Dose day and the next 3 days	Technique, mobility, zone-2 aerobic work, light isometrics; no heavy hypertrophy sessions	Highest drug exposure; the anabolic stimulus would be partly wasted
From day 4 to the next dose	Normal hypertrophy and strength volume; concentrate the highest-volume phases here	Low residual exposure
Injury, surgery or procedure	The physician decides whether to skip the dose	Rapamycin slows healing

- **CYP3A5 *1/*3 (expresser)**: clears more sirolimus per mg, so exposure tends to be lower. A sirolimus level 24–48 h after the dose calibrates real exposure. It is a narrow-therapeutic-index drug: the finding is **informational** and any adjustment belongs to the prescribing physician (Birdwell 2015; te Loo 2023).
- **Interactions**: concentrated polyphenols (curcumin, quercetin, fisetin, luteolin, apigenin) and grapefruit act on CYP3A4/P-gp and make exposure unpredictable; keep them out of the dose week.
- **Antioxidants**: NAC $\geq$ 6 h after training (suggestion, not applied), so as not to blunt adaptation signalling.
- **Longevity counterpoint**: FOXO3 G/G and rapamycin act on the same nutrient-sensing axis. Pulses of mTORC1 (training and protein) alternating with periods of inhibition (rapamycin) serve both goals.

17. Current management and genetic alignment: supplements

The real stack, at the doses being taken; nothing has been changed. Changes are **suggestions (not applied)**, for discussion with the physician. Fixed across the plan: zero alcohol; omega-3 3.2–4 g/day; melatonin 0.3–1 mg; D3 10.000 IU; 5-MTHF and never folic acid; ferritin < 100.

Supplement (current dose)	Relevant genes	Alignment	Comment
Creatine monohydrate 5 g	Lean mass p9.5–28; AMPD1, CKM; methylation	HIGHLY FAVOURABLE	The supplement with the most evidence for strength and lean mass (Kreider 2017; Desai 2024); spares methyl groups. Keep; kidney function by cystatin C.
Bovine collagen 20 g	COL5A1 T/T, COL1A1, MMP7	FAVOURABLE	Suggestion (not applied): take with ~50 mg vitamin C 30–60 min before loading, which raises collagen synthesis (Shaw 2017). Low in leucine: does not count towards muscle protein.
Fish oil (EPA 1125 + DHA 875 mg)	Recovery; SOD2; lean mass	FAVOURABLE	Omega-3 raises the anabolic response to amino acids (Smith 2011). Suggestion (not applied): 3.2–4 g/day EPA+DHA, EPA-rich.
D3 10.000 IU + K2 MK-7 240 µg	VDR Ff, GC T/T; genetic 25-OH-D p85	FAVOURABLE	Keep D3 at 10.000 IU: measured 25-OH-D stayed below target despite the dose. Vitamin D improves strength mainly when it is low (Beaudart 2014).
Magnesium glycinate 500 mg + acetyl-taurate 750 mg	Mg ²⁺ -ATP (kinases, ACSL1)	FAVOURABLE	ATP and sleep cofactor. Keep.
Glycine 3 g + TMG 1.5 g	Methylation (COMT, creatine synthesis); collagen	FAVOURABLE	Glycine for collagen and sleep; TMG supports methylation. Suggestion (not applied) from another analysis: TMG 3 g.
Trace minerals (zinc 11 mg, copper 0.9 mg, selenium, iodine, manganese, chromium, molybdenum, boron)	Zinc (protein synthesis, testosterone); copper (collagen lysyl oxidase)	FAVOURABLE	Zinc and copper measured low. Suggestion (not applied): + zinc ~10 mg + copper ~1 mg after caeruloplasmin; copper is a cofactor for collagen cross-linking.
Taurine 6 g	Muscle function	NEUTRAL	No demonstrated gain above 3 g. Suggestion (not applied) from another analysis: 3 g.
NAC 1800 mg (3 h after training)	SOD2 Ala/Ala; redox adaptation signalling	CAUTION	High-dose antioxidants close to training can blunt adaptations (Ristow 2009; Paulsen 2014a; Paulsen 2014b). Suggestion (not applied): 1200 mg at night, ≥ 6 h after training.
Liposomal vitamin C 500 mg	COL5A1 (collagen hydroxylation); HFE H63D	MIXED	Useful for tendon collagen if taken with collagen before loading; megadoses of vitamins C + E around training blunt adaptations. Always away from meals (iron; ferritin < 100).
Curcumin 1 g, luteolin 100 mg, apigenin 100 mg, fisetin 2.5 g + quercetin 250 mg (3×/month)	CYP3A5/CYP3A4 and P-gp (rapamycin)	CAUTION	They alter rapamycin exposure and are concentrated antioxidants. Suggestion (not applied) from another analysis: pause while ALT is high and never in the rapamycin week.
Pine bark extract 100 mg, hydroxytyrosol 25 mg + olive leaf	NOS3 ×2, EDN1	FAVOURABLE	Endothelial rationale (nitric oxide); weak performance evidence. Keep.
Psyllium 10 g + PHGG 6 g; tributyrin 500 mg; multi-strain probiotic	FTO, TCF7L2; body fat p92	FAVOURABLE	More fibre is always better (post-meal glucose, satiety). Suggestion (not applied): psyllium 15 g + PHGG 10 g.
Melatonin 0.3 mg (sometimes 100 mg)	MTNR1B C/G	MIXED	Sleep helps recovery. With MTNR1B, the dose stays at 0.3–1 mg, ≥ 3 h after the last meal; suggestion (not applied): drop the 100 mg.
Astaxanthin 12 mg, lycopene 35 mg, lutein 20 mg + zeaxanthin 4 mg	Oxidative stress	NEUTRAL	No demonstrated effect on strength gains; no conflict.
R-ALA 240 mg, ALCAR 500 mg, L-carnosine 2 g, ergothioneine 10 mg, Ca-AKG 2 g	Muscle energy	NEUTRAL	No demonstrated effect on hypertrophy; no conflict.
5-MTHF 800 µg, B12 1000 µg, B2 25 mg, B6 (P5P) 15 mg, lithium 5 mg	Methylation	NEUTRAL	No direct muscle effect. 5-MTHF, never folic acid. Suggestion (not applied) from another analysis: pause B6, which is high.
Aged garlic 1.2 g + black garlic 10 g, ginger 400 mg, sulforaphane	Recovery	NEUTRAL	No relevant training effect; no conflict.

18. Current management and genetic alignment: medications and peptides

Any drug change goes through the prescribing physician; pharmacogenomic findings are information for that conversation. LDL and ApoB: the lower, the better.

Drug (current dose)	Relevant genes	Alignment	Comment
Pitavastatin 4 mg	SLCO1B1 Val/Val; LILRB5 Asp/Gly	FAVOURABLE	No genetic myopathy risk (Cooper-DeHoff 2022) and no CYP3A4 (no rapamycin interaction). LILRB5: ask for CK if muscle pain appears (Siddiqui 2017).
Ezetimibe 10 mg	—	NEUTRAL	No muscle effect.
Bempedoic acid 180 mg	COL5A1 T/T	CAUTION	A tendon-rupture signal on the label (small, not significant in the large trial; Bays 2024) in a genetically stiffer tendon; slightly lowers haemoglobin. Suggestion (not applied) from another analysis: switch to inclisiran, which lowers LDL further.
Dapagliflozin 10 mg	TCF7L2; ketone bodies	MIXED	Favourable for heart and kidney. Long training in an energy deficit + SGLT2 inhibitor: rare risk of euglycaemic ketoacidosis; plenty of water and ketones if unwell.
Acarbose 200 mg 1–2×/day	TCF7L2, MTNR1B	NEUTRAL	Hypoglycaemia during training is corrected with dextrose, not table sugar. The above-label dose per meal is for discussion with the physician (liver).
Imeglimin 1000 mg 2×/day	—	NEUTRAL	No muscle data. Suggestion (not applied) from another analysis: stop.
Telmisartan 20 mg	AGTR1 A/C	NEUTRAL	AGTR1 is the drug target; no demonstrated training implication.
Tadalafil 5 mg/day	NOS3 ×2, EDN1	FAVOURABLE	Amplifies the nitric oxide–cGMP pathway, where the heterozygous variants sit; weak performance evidence. Nitrate concentrates only after talking to the physician.
Isotretinoin 40 mg/week	COL5A1 T/T; LILRB5	CAUTION	Vigorous training on isotretinoin can raise CK sharply (Marson 2022); joint and tendon pain are described. CK and ALT at check-ups; conservative load progression.
Rapamycin 4 mg every 6 weeks	MTOR G/T, FOXO3 G/G, CYP3A5 *1/*3, ABCB1	MIXED	Cuts training-induced protein synthesis in the days after the dose (Drummond 2009); the CYP3A5 expresser lowers exposure. See the timing section; level at 24–48 h (informational; narrow therapeutic index).
Low-dose naltrexone 4.5 mg	—	NEUTRAL	No muscle effect.
Retatrutide 8 mg every 6 days	Lean mass p9.5–28; body fat p92	MIXED	It targets fat (p92), but part of the loss is lean mass (about 1/4 of total loss with incretin agonists; Look 2025), on a genetically low lean mass. Progressive strength work, distributed protein, creatine and DEXA; the pace of the deficit is for discussion with the physician.

Peptides

Peptide (current schedule)	Alignment	Comment
MOTS-c 5 mg every 5 days × 21 days	NEUTRAL	"Exercise mimetic" only in animals; does not replace training.
SS-31 25 mg/day × 21 days	NEUTRAL	Mitochondria; no data on strength gains.
Oxytocin 160 µg	NEUTRAL	Muscle regeneration only in old mice.
5-amino-1MQ 100 mg	NEUTRAL	NNMT inhibitor; animal data only.
Epitalon 500 µg + pinealon 1 mg; humanin; thymosin α1; thymalin; FOXO4-DRI; glutathione 100 mg/week	NEUTRAL	No link to this pathway. Other analyses suggest dropping epitalon (suggestion, not applied).

19. Gaps and multivitamin

What the current stack lacks for this pathway, with priority and whether the multivitamin covers it. Everything is a **suggestion (not applied)**, for discussion with the physician.

Gap	Why (genes)	Suggestion (not applied)	Priority	Multivitamin covers it?
Collagen + vitamin C before loading (timing)	COL5A1 T/T, MPP7	15–20 g collagen or gelatin + ~50 mg vitamin C, 30–60 min before loading (uses the collagen already taken)	HIGH	Partly: has vitamin C, no collagen; timing is what matters
Distributed protein and a pre-sleep dose	Lean mass p9.5–28; IGF-1 p32; retatrutide	Keep ~140 g/day in 4 doses of 30–40 g, one before sleep (Morton 2018); do not raise the total before repeating urine albumin	HIGH	No
Tendon-specific load progression	COL5A1 T/T	Isometrics and heavy slow resistance; raise load ≤ 10% per week in exercises loading the Achilles and patellar tendons	HIGH	Not applicable
Omega-3 to 3.2–4 g/day	Recovery; anabolic response	3.2–4 g/day EPA+DHA, EPA-rich, with the fattiest meal	MODERATE	No
Zinc + copper	Zinc and copper measured low; lysyl oxidase	+ zinc ~10 mg + copper ~1 mg (bisglycinate), after caeruloplasmin	MODERATE	Yes: zinc 15 mg + copper 1 mg as bisglycinate
Training structure	Physical activity p7–14; BDNF Val/Met	Fixed slot in the diary, measured goals (strength, DEXA), training with company	MODERATE	Not applicable
Dietary nitrate	NOS3 ×2, EDN1	Leafy greens and beetroot with meals (Coggan 2021); concentrates only after talking to the physician (tadalafil)	LOW	No
Benfotiamine	B1 measured low; muscle energy (pyruvate dehydrogenase)	300 mg/day (proposed by another analysis)	LOW	No: the dose is far too low

Does the multivitamin cover the gaps?

The composition of the **reference multivitamin** (4 capsules a day) was checked ingredient by ingredient.

- **Covers:** zinc 15 mg + copper 1 mg as bisglycinate (the zinc and copper gap); vitamin C 400 mg (part of the collagen protocol); magnesium 150 mg (already covered).
- **Does not cover:** collagen, protein, omega-3, nitrate, or benfotiamine at a useful dose (it has 1 mg, plus 2.4 mg thiamine).
- **Conflicts with the current stack:** vitamin A 1500 µg partly preformed (retinyl palmitate), incompatible with isotretinoin, which requires zero preformed retinol; vitamin B6 (P5P) 6.7 mg with B6 already high; quercetin 50 mg + resveratrol 10 mg (concentrated polyphenols, rapamycin week); D3 2000 IU and K1 400 µg on top of D3 10.000 IU and K2; folate as 5-MTHF 680 µg (the right form, never folic acid) on top of the 800 µg; selenium 100 µg on top of the trace minerals.

Reading: in its current form, the multivitamin is not the best solution for this pathway. It only covers the zinc and copper gap and brings preformed vitamin A, B6 and polyphenols that conflict with the stack. A standalone zinc + copper does the same without those conflicts (suggestion, not applied).

20. Cross-references with prior reports

Prior report	What it said	What this report adds
Sports Profile: Endurance and Strength	Mixed profile, good aerobic engine, little baseline lean mass	Confirms with more genes and scores: lean mass p9.5–28 and PPARGC1A Gly/Gly.
Training Response and VO2max	No non-responder signal; weak spontaneous drive	Confirms: physical activity p7–14 and gym p12.6 (bottleneck 4).
Muscle, Bone and Sarcopenia	Genetically low muscle mass; favourable bone	Converges: low lean mass is the central weak point; COL1A1 G/G in the gene favours bone.
Tendons, Ligaments and Injuries	COL5A1 T/T as the only classic signal; typical scores	Gains importance: with strength as the priority, COL5A1 becomes the most unfavourable finding (bottleneck 1); muscle/ligament disorders p34.
Glucose, Insulin and Diabetes	Beta cell with little reserve; low average glucose	Converges: heterozygous TCF7L2 and MTNR1B with HbA1c p7.5; muscle is the glucose sink (bottleneck 3).
Nitric Oxide and Endothelium	NOS3 Glu298Asp and –786 T/C; EDN1	Converges: lower exercise vasodilation; tadalafil and dietary nitrate act on the same pathway.
Methylation, Homocysteine and Choline	High homocysteine; COMT Val/Val	Converges: creatine synthesis uses methyl groups, and supplemental creatine spares them; TMG serves both pathways.
Mitochondria and Energy	Favourable PPARGC1A and TFAM; SOD2 Ala/Ala	Confirms: good volume tolerance; high-dose antioxidants away from training.
Nutrient Sensing and Autophagy	mTOR/FOXO axis; rapamycin	Explicit trade-off: muscle needs mTORC1 pulses and longevity needs periods of inhibition; rapamycin timing reconciles both.

21. Link to the ageing mechanisms

Mechanisms from the longevity target list on which this genetic pathway bears.

Mechanism	Genes / scores	Direction	Lever
Sarcopenia; Loss of Lean Mass; Muscle Atrophy	Lean mass p9.5–28; IGF-1 p32	Unfavourable	Progressive strength work, distributed protein, creatine, DEXA
Loss of Muscle Strength	Grip p51; ACTN3 RX	Typical	Strength and power training
Frailty	Frailty factor p75	Mildly unfavourable	Strength, power and balance
Neuromuscular Junction Dysfunction	Normal CNTF; BDNF Val/Met	Typical	Power and technique training
Osteoporosis	COL1A1 G/G in the gene	Favourable	Progressive impact loading
Extracellular Matrix Remodelling and Stiffness	COL5A1 T/T, MMP7	Unfavourable	Collagen + vitamin C before loading; slow progression
Deregulated Nutrient Sensing; Autophagy	MTOR G/T, FOXO3 G/G	Favourable	mTORC1 pulses alternating with rapamycin
Mitochondrial Dysfunction; Decline of Oxidative Phosphorylation	PPARGC1A Gly/Gly, TFAM	Favourable	Zone-2 aerobic work and volume
Oxidative Stress; Redox Imbalance	SOD2 Ala/Ala	Mixed	High-dose antioxidants away from training
Chronic Inflammation; Inflammaging	TNF, CCL2; CRP p26	Favourable	Omega-3 3.2–4 g; sleep
Insulin Resistance; Hyperglycaemia and Glucose Intolerance	TCF7L2, MTNR1B; HbA1c p7.5	Mixed	Muscle as a sink; more fibre; melatonin 0.3–1 mg
Adipose Tissue Dysfunction; Decline of Metabolic Homeostasis	Body fat p92; FTO	Unfavourable	Strength + aerobic work; lean mass protected on retatrutide
Endocrine Dysregulation	SHBG p83; FAM9B T; total testosterone p70	Mildly unfavourable	Measure free testosterone and SHBG
Vascular Ageing; Arterial Stiffness	NOS3 ×2, EDN1, AGTR1	Mildly unfavourable	Aerobic work; dietary nitrate; low LDL/ApoB
Altered Intercellular Communication (myokines)	IL6 G/G, IL15RA A/A	Typical	Regular muscle contraction
Stem Cell Exhaustion (satellite cells)	IGF1; typical myostatin and follistatin	Typical	Regular strength training
Sleep Disruption	MTNR1B C/G	Mildly unfavourable	Melatonin 0.3–1 mg; regular schedule

22. Suggested monitoring panel

For discussion with the physician. Calendar: **Now** (next blood draw or visit), **Coming years** (periodic reassessment), **Later** (if symptoms appear or new technology).

Test	Why	When
DEXA (appendicular lean mass and fat)	Lean mass p9.5–28; body fat p92; retatrutide	Now
Grip strength (dynamometer) and baseline strength test	Grip p51; track the real training response	Now
Resting CK (≥ 48 h without training) with ALT and AST	LILRB5; isotretinoin; statin; separates muscle from liver origin	Now
Total testosterone, SHBG, calculated free testosterone, LH, oestradiol	SHBG p83; FAM9B T	Now
IGF-1	Genetic IGF-1 p32	Now
Blood sirolimus 24–48 h after the dose	CYP3A5 expresser (informational)	Now
25-OH-D and calcium (with PTH)	VDR, GC; D3 10.000 IU	Now
Ferritin, transferrin saturation, full blood count	HFE H63D; genetic ferritin p80; target < 100	Now
HbA1c or fructosamine, fasting insulin	TCF7L2, MTNR1B	Now
ApoB and LDL	The lower, the better	Now
Omega-3 index	Recovery; 3.2–4 g dose	Now
Thiamine (TDP) and cystatin C	Muscle energy; creatine raises creatinine	Now
VO2max and heart-rate recovery	PPARGC1A; aerobic response	Coming years
Repeat DEXA and baseline strength	Track lean mass over time	Coming years
Achilles or patellar tendon ultrasound	COL5A1 T/T; only if painful	Later
Sequencing (AR CAG repeat, ACE insertion/deletion, KANSL1)	Not readable on the chip	Later

23. Bibliography

Peer-reviewed PubMed-indexed sources and public scientific databases cited in this report. Format: author, year, journal, PMID, topic.

- **Bays HE et al.** (2024). *J Clin Lipidol*. PMID 37951797. safety of bempedoic acid.
- **Beaudart C et al.** (2014). *J Clin Endocrinol Metab*. PMID 25033068. meta-analysis: vitamin D and strength.
- **Birdwell KA et al.** (2015). *Clin Pharmacol Ther*. PMID 25801146. CPIC guideline CYP3A5 and tacrolimus.
- **Browning BL et al.** (2018). *Am J Hum Genet*. PMID 30100085. genotype imputation method.
- **Clarkson PM et al.** (2005). *J Appl Physiol*. PMID 15718405. ACTN3 and strength gains with training.
- **Coggan AR et al.** (2021). *J Int Soc Sports Nutr*. PMID 34625064. dietary nitrate and muscle power.
- **Cooper-DeHoff RM et al.** (2022). *Clin Pharmacol Ther*. PMID 35152405. CPIC guideline SLCO1B1/ABCG2/CYP2C9 and statins.
- **Desai I et al.** (2024). *J Strength Cond Res*. PMID 39074168. meta-analysis: creatine and body composition.
- **Drummond MJ et al.** (2009). *J Physiol*. PMID 19188252. rapamycin blocks contraction-induced protein synthesis.
- **Kilpeläinen TO et al.** (2011). *PLoS Med*. PMID 22069379. physical activity attenuates FTO.
- **Kreider RB et al.** (2017). *J Int Soc Sports Nutr*. PMID 28615996. ISSN position stand on creatine.
- **Lambert SA et al.** (2021). *Nat Genet*. PMID 33692568. PGS Catalog.
- **Landrum MJ et al.** (2018). *Nucleic Acids Res*. PMID 29165669. ClinVar.
- **Look M et al.** (2025). *Diabetes Obes Metab*. PMID 39996356. body composition with an incretin agonist.
- **Ma F et al.** (2013). *PLoS One*. PMID 23358679. ACE/ACTN3 and performance meta-analysis.
- **Mahalati K, Kahan BD** (2001). *Clin Pharmacokinet*. PMID 11523724. clinical pharmacokinetics of sirolimus.
- **Marson JW et al.** (2022). *JAAD Case Rep*. PMID 35242965. isotretinoin, exercise and CK.
- **Moel M et al.** (2025). *Aging (Albany NY)*. PMID 40188830. PEARL trial: intermittent rapamycin for one year.
- **Morton RW et al.** (2018). *Br J Sports Med*. PMID 28698222. meta-analysis: protein and resistance training.
- **Murphy C, Koehler K** (2022). *Scand J Med Sci Sports*. PMID 34623696. energy deficit and lean-mass gains.
- **Paulsen G et al.** (2014). *J Physiol*. PMID 24492839. vitamins C and E and endurance adaptation.
- **Paulsen G et al.** (2014). *J Physiol*. PMID 25384788. vitamins C and E and signalling after strength training.
- **Puthuchear Z et al.** (2011). *Sports Med*. PMID 21615186. review: ACE gene and performance.
- **Ristow M et al.** (2009). *Proc Natl Acad Sci USA*. PMID 19433800. antioxidants prevent exercise benefits.
- **Schoenfeld BJ et al.** (2017). *J Sports Sci*. PMID 27433992. weekly volume dose-response and hypertrophy.
- **Shaw G et al.** (2017). *Am J Clin Nutr*. PMID 27852613. vitamin C-enriched gelatin and collagen synthesis.
- **Siddiqui MK et al.** (2017). *Eur Heart J*. PMID 29020356. LILRB5 and statin intolerance.
- **Smith GI et al.** (2011). *Am J Clin Nutr*. PMID 21159787. omega-3 and muscle protein synthesis.
- **Sollis E et al.** (2023). *Nucleic Acids Res*. PMID 36350656. GWAS Catalog.
- **te Loo DMW et al.** (2023). *Pharmacogenomics*. PMID 37551646. systematic review: genetics and sirolimus pharmacokinetics.
- **Whirl-Carrillo M et al.** (2021). *Clin Pharmacol Ther*. PMID 34216021. PharmGKB evidence levels.
- **Willems SM et al.** (2017). *Nat Commun*. PMID 29313844. grip strength GWAS.